Neural Networks 142 (2021) 375–387

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Self-spectral learning with GAN based spectral–spatial target detection for hyperspectral image

Weiying Xie [a], Jiaqing Zhang [a,*], Jie Lei [a,*], Yunsong Li [a], Xiuping Jia [b]

[a] State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China
[b] School of Engineering and Information Technology, The University of New South Wales, Canberra, ACT 2600, Australia

ARTICLE INFO

Article history:
Received 19 July 2020
Received in revised form 24 March 2021
Accepted 26 May 2021
Available online 4 June 2021

Keywords:
Spatial–spectral target detection
Self-spectral learning
Feature extraction
Band selection
Hyperspectral image (HSI)

ABSTRACT

To alleviate the shortcomings of target detection in only one aspect and reduce redundant information among adjacent bands, we propose a spectral–spatial target detection (SSTD) framework in deep latent space based on self-spectral learning (SSL) with a spectral generative adversarial network (GAN). The concept of SSL is introduced into hyperspectral feature extraction in an unsupervised fashion with the purpose of background suppression and target saliency. In particular, a novel structure-to-structure selection rule that takes full account of the structure, contrast, and luminance similarity is established to interpret the mapping relationship between the latent spectral feature space and the original spectral band space, to generate the optimal spectral band subset without any prior knowledge. Finally, the comprehensive result is achieved by nonlinearly combining the spatial detection on the fused latent features with the spectral detection on the selected band subset and the corresponding selected target signature. This paper paves a novel self-spectral learning way for hyperspectral target detection and identifies sensitive bands for specific targets in practice. Comparative analyses demonstrate that the proposed SSTD method presents superior detection performance compared with CSCR, ACE, CEM, hCEM, and ECEM.

© 2021 Elsevier Ltd. All rights reserved.

1. Introduction

Hyperspectral image (HSI) processing techniques have attracted much interest in earth observation and deep space exploration since HSIs can offer intrinsic properties of different materials in the form of hundreds of adjacent spectral channels (Yang & Shi, 2016). In particular, target detection in HSIs aims to distinguish the target of interest from a large number of complex background classes, which is a common and practical need (Zhai, Liu, & Sun, 2019).

In the past decades, a variety of algorithms for target detection in HSI have emerged to overcome the above difficulties. Spectral angle mapper (SAM) (Kruse et al., 1993a) and spectral information divergence (SID) (Chang, 2000) are representative of traditional mainstream algorithms that measure the spectral distance between the test pixel and the prior target spectral signature without any assumption on data distribution. The adaptive coherence estimator (ACE) (Kraut, Scharf, & Butler, 2005), matched filter (MF) (Lockwood, Cooley, Jacobson, & Manolakis, 2009), and matched subspace detector (MSD) (Scharf & Friedlander, 1993) are derived from a generalized likelihood ratio test (GLRT) (Kelly, 1986), which regard the target detection task as hypothesis test problems depending on Gaussian distribution. Some variants based on MF, such as adaptive MF (AMF) (Robey, Fuhrmann, Kelly, & Nitzberg, 1992) and spectral MF (SMF) (Nasrabadi, 2008), have been successfully used in target detection. The orthogonal subspace projection (OSP) (Harsanyi & Chang, 1994), adaptive subspace detector (ASD) (Kraut, Scharf, & McWhorter, 2001), MI-SMF and MI-ACE (Zare, Jiao, & Glenn, 2018) are in allusion to subpixel. The typical emerging detector is constrained energy minimization (CEM) (Farrand & Harsanyi, 1997) that builds a finite impulse response (FIR) filter by imposing a constraint on the target or background to minimize the output energy. Subsequently, numerous variants of the CEM detector were presented such as hierarchical CEM (hCEM) (Zou & Shi, 2016) and ensemble-based cascaded CEM (ECEM) (Zhao, Shi, Zou, & Zhang, 2019). The hCEM exploits a nonlinear suppression function to suppress background through hierarchical operations, thereby improving detection performance. Not content with a mixture of the background and target pixels, sparse representation (SR) (Chen, Nasrabadi, & Tran, 2011) based target detection algorithms have been proposed in recent years, which is inspired by the observation that pixels in HSIs can only be represented by a few elements in the basis or dictionary (Charles, Olshausen, & Rozell, 2011). For instance, the combined sparse and collaborative representation (CSCR) (Li, Du, & Zhang, 2015) and Sparse and

* Corresponding authors.
E-mail addresses: jqzhang_2@stu.xidian.edu.cn (J. Zhang), jielei@mail.xidian.edu.cn (J. Lei).

https://doi.org/10.1016/j.neunet.2021.05.029
0893-6080/© 2021 Elsevier Ltd. All rights reserved.

Low-Rank Matrix Decomposition (Bitar, Cheong, & Ovarlez, 2019) achieve target detection by evaluating the difference between the two representation residuals of the background and target.

Unfortunately, a great spectral similarity that appears between target and background, and vast amounts of spectral information that is highly correlated among bands make the hyperspectral target detection task be a challenging one. Besides, the targets always occupy a few pixels in the entire image while the background owns various land cover classes, which include almost all the pixels (Manolakis & Shaw, 2002). Because of the variety and complexity of the background, there may be a background similar to the spectral vector of the target, which hinders the detection of the spectral detector (Zhang, Zhang, Tao, Huang, & Du, 2014). The severe imbalance multiplies the difficulties to obtain discriminative features between targets and the background from the complex HSIs.

Nowadays, deep neural networks (DNNs), like autoencoder (AE) (Hinton & Zemel, 1994), adversarial AE (AAE) (Makhzani, Shlens, Jaitly, Goodfellow, & Frey, 2015), generative adversarial network (GAN) (Goodfellow et al., 2014) and convolutional neural network (CNN) (Gu et al., 2015), have demonstrated outstanding effect on digging out intrinsic and discriminative features from the high-dimensional and complex HSIs (Chen, Jiang, Li, Jia, & Ghamisi, 2016; Ma, Wang, & Geng, 2016; Xie, Yang, Lei, Li, Du, & He, 2020; Zhang, Gong, Mao, Li, & Wu, 2019). Based on the fact that the capabilities of both nonlinear discrimination and generalization are significant for hyperspectral target detection (Zhao et al., 2019), DNNs have been successfully applied for hyperspectral fields (Jiang, Li, Xie, & Du, 2020; Li, Wu, & Du, 2017; Liu et al., 2019). For the strong representational capability of DNNs with the cascaded structure and nonlinear function, the target detection algorithms based on DNNs are more suitable for the complex scene in HSIs. Some work of band selection based on DNNs has been carried out, like Cai, Liu, and Cai (2020), Zhan et al. (2017), in which the physical significance of spectral signatures can be preserved, and the saliency bands can be used for further classification (Cai et al., 2020; Feng, Li, Chen, Zhang, Tang, & Wu, 2019) or detection (Xie, Lei, Yang, Yang, Du & Li, 2020; Xie, Li, Lei, Yang, Chang, & Li, 2020). The redundant information and noise interference are effectively removed and thus have a positive influence on HSI application to a certain degree. Typically, deep latent spectral representation learning (DLSRL) was originally proposed for hyperspectral band selection, followed by the target detection task, which has proven to have a positive impact on detection performance and computational complexity. Therefore, the great spectral similarity that appears between target and background makes it particularly important to combine with discriminative spatial features for target detection.

To maintain comprehensive detection of spatial and spectral regions while releasing redundancy and reducing computational burden, we bring forward a SSTD method based on SSL with GAN for the first time in the field of hyperspectral target detection. Speaking of the SSL, it mainly considers our issue of imbalanced data and insufficient target spectral samples, which are different from supervised learning with sufficient training samples (Badrinarayanan, Kendall, & Cipolla, 2015; Redmon & Farhadi, 2018). In particular, GAN is adopted to achieve SSL in SSTD since GAN is particularly prominent with its powerful generalization capability and can learn useful representation features for unlabeled data in a low-dimensional space. More importantly, based on SSL, an innovative band selection rule is introduced to determine the physically meaningful subset of hyperspectral bands with the target–background distinguishing feature. Subsequently, spectral detection based on subset and spatial detection is conducted simultaneously. A nonlinear fusion framework combines the two to alleviate unsatisfactory detection performance caused by poor structural information in the spatial domain and mixed spectral signature between the target and the background. As a result, the box plots of the real HSIs indicate that the selected band subset based on SSL with GAN can be more powerful to stand out discriminative features and strengthen separation gaps between the target and background. In addition to band selection, experiments also show that the combination of spatial and spectral modules for hyperspectral target detection serves two advantages: not only does it achieve better performance, it also offers an insight into the bands where the target response is extremely sensitive.

Compared with other existing hyperspectral target detection algorithms, the major contributions of our SSTD method are listed as follows.

- A novel spectral–spatial detection framework in deep latent space is designed to alleviate the shortcomings of target detection in only one aspect, thus offering a comprehensive result with detection ability maintained and false alarm rate reduced. Especially, the spatial detection is completely unsupervised, and the spectral detection exploits the selected target signature based on the optimal rule.
- The concept of SSL by GAN is introduced into hyperspectral band selection in an unsupervised fashion for the first time, which not only relieves the computational burden and redundant information without any prior knowledge but also offers an insight into the bands with background suppression and target saliency. Moreover, the architecture of GAN is specially designed to avoid generating out-of-classes spectra.
- To offer an effective mapping relationship between the latent spectral feature space and the original spectral band space, an innovative structure-to-structure selection rule that takes full account of the mean, variance, and covariance is established. Experiments demonstrate that the proposed method can identify sensitive bands for specific targets such as airports, which naturally provides great feasibility for practical applications.

The remainder of this paper is organized as follows. Section 2 elaborates on the proposed SSTD framework based on the selected subset offered by SSL in detail. Section 3 demonstrates the evaluation experiments. Finally, Section 4 concludes the paper.

2. Target detection framework

Let $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_i, \ldots \mathbf{X}_{M\times N}] \in \mathbb{R}^{M\times N\times L}$ represent an input HSI with $M \times N$ pixels and L spectral bands, where $\mathbf{X}_i$ represents the ith spectral vector, meanwhile $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{i=L}$, where $\mathbf{x}_i$ is the ith band of $\mathbf{X}$. Note that all values in $\mathbf{X}$ are within the range of (0, 1) through normalization.

The traditional methods treat the hyperspectral target detection task as an optimal spectral matching problem with the known target spectral signature so that they are always carried out in a single spectral dimension with huge data volumes. Although the targets in HSIs are a small fraction of the scene and relatively poorly structured relative to the background, they still have spatially specific properties that are worth to be explored. To achieve a comprehensive result based on a lower-dimensional data volume, we mathematically define and formulate our problem as follows:

$$\mathbf{T} = \beta_1 \{\mathbf{T}_1 | f(\mathbf{Z}; \mathbf{X})\} + \beta_2 \left\{\mathbf{T}_2 \middle| \left(\hat{\mathbf{d}}, \mathbf{Y}\right) \leftarrow f(\mathbf{Z}; \mathbf{X})\right\} \tag{1}$$

where β_1 and β_2 are the weighting factors corresponding to the spatial detection $\mathbf{T}_1$ and spectral detection $\mathbf{T}_2$, respectively. The function $f(\mathbf{Z}; \mathbf{X})$ in Eq. (1) denotes the proposed self spectral-learning (SSL) module through GAN. It aims to map the input HSI $\mathbf{X}$ into the deep latent space $\mathbf{Z}$ in which the dimension can be

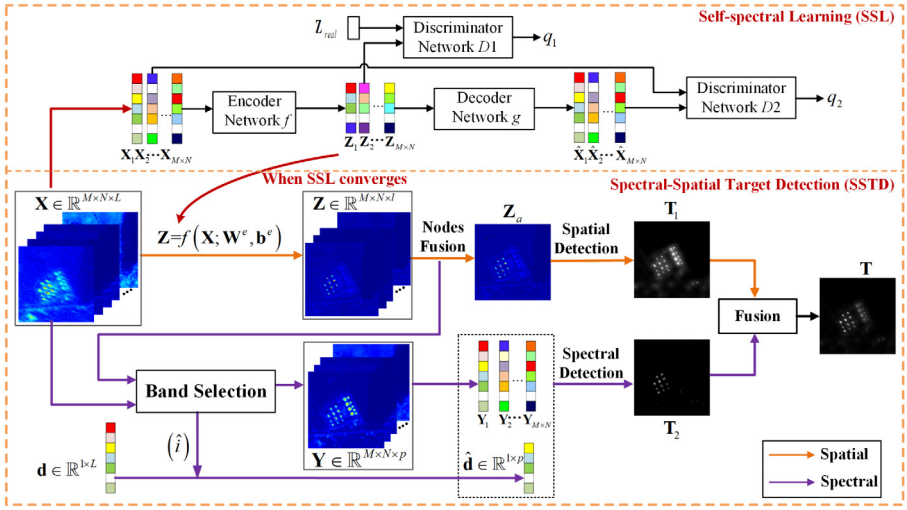

Fig. 1. Schematic of the proposed SSTD framework in HSI.

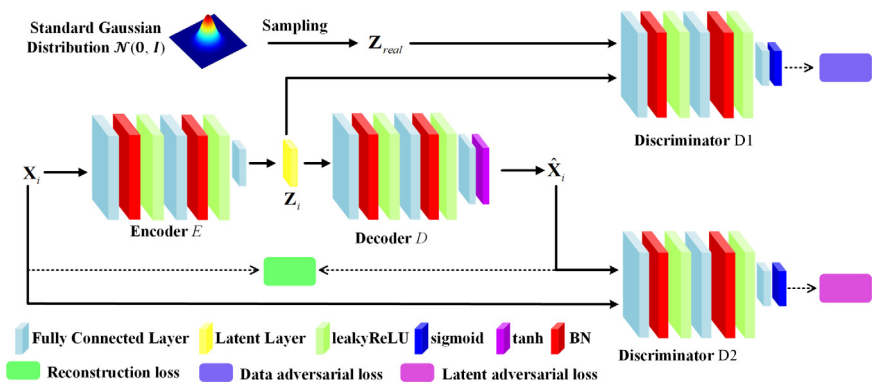

Fig. 2. Details of reconstruction loss and adversarial learning functions based on SSL.

reduced with discriminative information preserved. On the one hand, the spatial detection $\mathbf{T}_1$ relies entirely on a self learning process f ($\mathbf{Z}$; $\mathbf{X}$), and does not require any prior information. Thereby the problem of insufficient training samples in hyperspectral target detection is overcome. On the other hand, the spectral detection $\mathbf{T}_2$ is performed on the selected band $\mathbf{Y}$ obtained by an innovative structure-to-structure selection rule between the original band space and deep latent space. Correspondingly, the known target spectral signature $\mathbf{d}$ is also selected as $\hat{\mathbf{d}}$ based on the optimization criterion. Fig. 1 shows a schematic of the proposed SSTD framework. It is composed of three major components: SSL f ($\mathbf{Z}$; $\mathbf{X}$), spatial detection in deep latent space $\mathbf{T}_1$ and spectral detection on band subset $\mathbf{T}_2$. The solving process of these components is explained in detail in the following subsections.

2.1. Self-spectral learning (SSL)

Contrary to natural images, the amount of data generated by hyperspectral imaging sensors is quite huge. In addition to the vast amount of data information, hyperspectral target detection suffers from the curse of dimensionality. We argue that if the entire latent space is limited to characterize HSIs, the reconstruction error of HSIs will be minimal, thereby generating latent features that can fully represent the original data. Hence, we explicitly model projections of the observed HSI into the entirety of the latent space with the SSL module.

A spectral GAN is established to conduct SSL, as shown in Fig. 2 including encoder, decoder, and two discriminators. The encoder learns to map the input spectral vector to the underlying code, and the decoder reconstructs the input data from the hidden layer. Then the discriminator D_1 is trained to distinguish whether its input is generated by the encoder or sampled from the prior distribution specified by the user. At the same time, the discriminator D_2 tries to distinguish between the original input spectral vector and the reconstruction one. The spectral mapping encoding with the function f and decoding with function g can be defined as:

$$\mathbf{Z}_i = f(\mathbf{W}^e\mathbf{X}_i + \mathbf{b}^e) \tag{2}$$

$$\hat{\mathbf{X}}_i = g(\mathbf{W}^d\mathbf{Z}_i + \mathbf{b}^d) \tag{3}$$

where the parameters $\mathbf{W}^e$ and $\mathbf{b}^e$ represent the trainable weight matrix and bias vector of encoding process and the parameters $\mathbf{W}^d$ and $\mathbf{b}^d$ are for decoding process, respectively. The encoder and decoder are trained using mean squared error loss as:

$$L_r = \|\mathbf{X}_i - \hat{\mathbf{X}}_i\|_2^2 \tag{4}$$

Besides, adversarial learning terms in the latent feature space and data space described in the following subsections are also adopted to learn parameters of the GAN.

2.1.1. Adversarial learning in latent feature space

In this subsection, the encoder E is regarded as a generator to obtain hidden units $\mathbf{Z}$, i.e., false samples, which are sent into the discriminator D_1. The true samples $\mathbf{Z}_{real}$ follow a standard

Gaussian distribution $N(0, 1)$, as another input for D_1. The generator E is trained to maximally confuse the discriminator into believing that the generated samples come from the true samples. The discriminator D_1 tries to distinguish the true samples from false samples. Hence, the latent adversarial loss between E and D_1 can be expressed as follows:

$$L_z = \min_E \max_{D_1} \mathrm{E}_{\mathbf{Z}_{real}\sim N(0,1)} [\log D_1(\mathbf{Z}_{real})] + \mathrm{E}_{\mathbf{Z}_i\sim p(\mathbf{Z}_i)} [\log(1 - D_1(E(\mathbf{X}_i)))] \tag{5}$$

2.1.2. Adversarial learning in data space

To reject out-of-class features produced by the previous part, the discriminator D_2 is introduced. The main function of the autoencoder including encoder and decoder considered as a generator G in this part is to reconstruct the input spectral vector to fool the discriminator D_2 while the discriminator attempts to compute the probability of the real input image $\mathbf{X}$ rather than the generated data $\hat{\mathbf{X}}$. The generator learns the characteristic of real spectral vectors by minimizing the difference from $\hat{\mathbf{X}}$ to $\mathbf{X}$. To avoid being fooled by G, D_2 tries to maximize the discernibility to tell whether the sample is from the generator or the original input. During this period, D_2 is helpful for the generator to obtain more stable and robust parameters in the training process. We drive latent adversarial loss to express this optimization problem as:

$$L_x = \min_G \max_{D_2} \mathrm{E}_{\mathbf{X}_i\sim p_{data}} [\log D_2(\mathbf{X}_i)] + \mathrm{E}_{\mathbf{Z}_i\sim p(\mathbf{Z}_i)} [\log(1 - D_2(D(\mathbf{Z}_i)))] \tag{6}$$

Through the above adversarial training process and optimization, the total loss function including three components, the reconstruction loss L_r, the latent adversarial loss L_z, and the data adversarial loss L_x, which can be formulated as:

$$L = \alpha_0 L_r + \alpha_1 L_z + \alpha_2 L_x \tag{7}$$

where α_0, α_1, α_2 are the weighting parameters to balance the effect of each individual part on the entire calculation.

2.1.3. Network architecture

Each pixel in the spatial domain corresponding to a spectral vector is treated as a training sample, and the training batch size is equal to the number of spatial pixels in an HSI, i.e., $M \times N$. A fully connected network is employed with four layers input-middle-middle-output, and the units of all layers in different components can be described as $E\{L, 500, 500, l\}$, $D\{l, 500, 500, L\}$, $D_1\{l, 500, 500, 1\}$ and $D_2\{L, 500, 500, 1\}$, respectively. Here the variate l is the number of hidden units in the deep latent layer restricted to be $\sqrt{L} + 1$ (Cao, Nicolau, & Mcdermott, 2016) in practice. All fully connected layers in the SSL network employed *LeakyReLU* (with slope 0.2) as activation function, while the last layer of encoder has no activation function and the last layer of decoder employs *tanh* and the last layer of two discriminators uses *sigmoid* as shown in Fig. 2. The distribution of input in each layer changes during training, as the parameters of the previous layer change and training in deep neural networks becomes complicated. This problem is referred as internal covariate shift which can be addressed by batch normalization (BN) (Ioffe & Szegedy, 2015), so BN is utilized before the *LeakyReLU* in E, D, $D1$ and $D2$ to generate activation with a stable distribution and accelerate the training speed. Moreover, the expression of $BN(x)$ is as follows:

$$BN(x) = \frac{x - E(x)}{\sqrt{var[x] + \varepsilon}} \tag{8}$$

where $E(x)$ and $var[x]$ represent the mean and variance of mini-batch, respectively. Here, ε is set to 10^{-8} to avoid zero denominators and x is the input of each fully connected layer in our architecture. All structures are trained by a stochastic Adam optimizer with learning rate set to 0.001 for E and D and 0.0001 for D_1 and D_2, respectively. Training is stopped after 10000 epochs. The complete training network architecture is shown clearly in Fig. 2. When the training converges and then optimal solution is obtained, the term $f(\mathbf{Z}; \mathbf{X})$ is transformed to $\mathbf{Z} = f(\mathbf{X}; \mathbf{W}^e, \mathbf{b}^e)$, where the weight $\mathbf{W}^e$ and bias $\mathbf{b}^e$ of encoder are fixed.

2.2. Spatial detection in deep latent space

The nodes of hidden layer are fused to remove noisy pixels, discard redundant information and extract useful spatial information in spatial detection. In addition, the fusion map is regarded as reference for latter band selection in spectral detection. Generally speaking, average fusion is widely used in various fields and it is based on the theory that all of bands are in the equal importance. However, in fact, not all bands provide the same information content. Thus, we adopt an adaptive fusion (Xie, Jiang, Li, Jia, & Lei, 2019) to complete the combination of the hidden nodes in consideration of avoiding the drawbacks in average fusion:

$$\mathbf{Z}_a = \sum_{k=1}^{l} \lambda_k \mathbf{Z}_k \tag{9}$$

where the weight coefficient for each hidden feature matrix $\mathbf{Z}_k$ is defined as $[\lambda_1, \lambda_2, \ldots, \lambda_l]^T$. The structure tensor of kth hidden unit $\mathbf{Z}_k$ at pixel i can be decomposed by:

$$T_i^k = \begin{bmatrix} \eta_i^k & \gamma_i^k \end{bmatrix} \begin{bmatrix} \mu_i^k & 0 \\ 0 & \sigma_i^k \end{bmatrix} \begin{bmatrix} \eta_i^k \\ \gamma_i^k \end{bmatrix} \tag{10}$$

where μ_i^k and σ_i^k are the corresponding non-negative eigenvalues of kth hidden node at pixel i. μ_i^k is the larger eigenvalue which illustrates the edge intensity of $\mathbf{Z}_i^k$ and the corresponding eigenvector, η_i^k, indicates the orientation that maximizes the spectral value fluctuations. The weight coefficient for each feature map at pixel i can be calculated by:

$$\lambda_i^k = \frac{\mu_i^k}{\sum_{k=1}^{l} \mu_i^k} \tag{11}$$

where λ_i^k is the weight of the kth hidden unit at pixel i. By the adaptive fusion, we can not only get a comprehensive feature map for spatial detection but also obtain the reference image for band selection in spectral detection.

Since targets in HSIs always appear in a small area and have high contrast compared with the background, morphological attribute filter (Dalla Mura, Atli Benediktsson, Waske, & Bruzzone, 2010) is utilized to improve detection performance. The bright connected components are eliminated by the attribute opening operation $O(\mathbf{Z}_a)$ (Breen & Jones, 1996). Therefore, the connected components with a small area target are preserved in hidden feature map $\mathbf{Z}_a$ by:

$$\mathbf{D}_a = |\mathbf{Z}_a - O(\mathbf{Z}_a)| \tag{12}$$

In consideration of the local smooth prior and reduction of false alarm rate, guided filter (Xie et al., 2019) is applied to refine the fused image. The final spatial map is given by:

$$\mathbf{T}_2 = GF(\mathbf{D}_a) \tag{13}$$

The filter size and blur degree of guided filter are set to 3 and 0.5 (Xie et al., 2019) for all datasets, respectively.

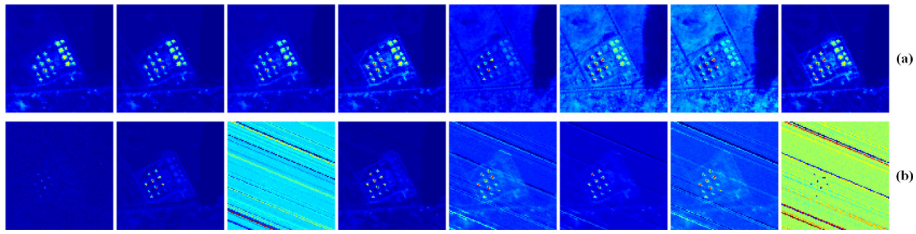

Fig. 3. The result of band selection in Texas Coast-2 : (a) some selected bands (b) some deleted bands.

2.3. Spectral detection in deep latent space

Since target detection in the original HSI space suffers from the curse of dimensionality, we present a structure-to-structure selection rule based on the latent feature matrix. The structural similarity index (SSIM) (Wang, Bovik, Sheikh, & Simoncelli, 2004) is an image quality evaluation criterion, which measures the structure, contrast, and luminance similarity between the original image and the processed image. Since SSL can get the full representation of original HSIs and the distance between targets and background in latent space is fully separable, the fused node of hidden feature matrix, $\mathbf{Z_a}$, is employed as a reference image. We compare each band of the original HSI with the reference image through the SSIM value. Note that the higher the value of SSIM, the greater the similarity between the two images. Consequently, the selection principle is constructed as:

$$s = \{s_i\}_{i=1}^{i=L} = \{\underbrace{s_m, s_n, \ldots, s_q}_{p}, \ldots, s_r\} \tag{14}$$

Here, we can select p bands which contain enough information to distinguish target from background and the remain is removed. s_i represents the SSIM evaluation:

$$s_i = SSIM\ (\mathbf{x}_i, \mathbf{Z}_a) \tag{15}$$

The number of bands p is flexible and can be set as ideal size by the following equation:

$$p = \varepsilon \cdot L \tag{16}$$

where L is the number of the original HSI and ε controls the balance between removing redundant information and retaining target information. Given reference image $\mathbf{Z}_a$, the optimal sequence number $\hat{i}$ of the selected band subset $\mathbf{Y}$ and the corresponding prior-known spectral vector $\hat{d}$ are defined as:

$$\mathbf{Y} = \{\mathbf{x}_i | i \in \hat{i} = \{m, n, \ldots, q\}\} \tag{17}$$

$$\hat{\mathbf{d}} = \{\mathbf{d}_i | i \in \hat{i} = \{m, n, \ldots, q\}\} \tag{18}$$

The pseudocode of the proposed band selection algorithm is described in Algorithm 1. As shown in Figs. 3 and 4, the band selection, an unsupervised method, based on a structure-to-structure rule can screen out the informative bands, eliminate the noisy, low-quality, and low-contrast bands.

The CEM detector (Farrand & Harsanyi, 1997) aims to enhance the output of target spectrum while refraining the energy of background spectrum in order to distinguish target from background. We develop CEM for the sake of improving further accuracy:

$$\mathbf{T}_2 = \frac{\hat{\mathbf{d}}\mathbf{R}^{-1}}{\hat{\mathbf{d}}\mathbf{R}^{-1}\hat{\mathbf{d}}^T}\mathbf{Y}^T \tag{19}$$

As illustrated in Fig. 1, the input of advanced CEM spectral detector is the optimal selected band subset $\mathbf{Y} \in \mathbb{R}^{C\times p}$ and the corresponding prior-known spectral vector $\hat{\mathbf{d}} \in \mathbb{R}^{1\times p}$ representing the target spectrum of interest. Here, the number of pixels C equals the length M multiplied by the width N of HSIs and p refers to the number of the selected bands. $\mathbf{R} \in \mathbb{R}^{p\times p}$ is the maximum likelihood estimation of the correlation matrix of the spectral data:

$$\mathbf{R} = \frac{1}{N}\mathbf{Y}^{\mathrm{T}}\mathbf{Y} \tag{20}$$

At this point, two initial maps are provided according to spatial and spectral information, respectively. The final comprehensive map is regulated by a spatial–spectral joint target detection as:

$$\mathbf{S} = \beta_1\mathbf{T}_1 + \beta_2\mathbf{T_2} \tag{21}$$

where β_1 and β_2 are also derived from adaptive fusion to distinguish whether the spatial feature or spectral feature is more important for the specific data. As shown in Fig. 1, the initial map $\mathbf{T}_1$ in the spacial domain always appears with low false alarm and properly background suppression. Thus, an exponential constrained nonlinear transform function based on $\mathbf{T}_1$ is introduced to suppress the background and enhance the detection performance:

$$\chi = 1 - e^{-\delta\mathbf{T_1}} \tag{22}$$

where δ is a positive parameter to adjust the degree of inhibition. The final map is calculated as:

$$\mathbf{T} = \chi \cdot \mathbf{S} \tag{23}$$

All in all, a remarkable algorithm is recommended to avoid the limitation of only detecting in a single perspective for HSI with tremendous data volume.

Algorithm 1 Realization of Band Selection

Input: original HSI $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{i=L}, \mathbf{x}_i \in \mathbb{R}^{M\times N}$, feature map $\mathbf{Z}_a \in \mathbb{R}^{M\times N}$, the number of bands L, controlling parameter ε, prior-known target spectral signature $\mathbf{d} \in \mathbb{R}^{\mathbf{1}\times\mathbf{L}}$.
Output: selected band subset $\mathbf{Y} \in \mathbb{R}^{M\times N\times p}$, selected target spectral signature $\hat{\mathbf{d}} \in \mathbb{R}^{1\times p}$.
Steps:

1: Calculate the SSIM values according to Eq. (15) for each $\mathbf{x}_i$ with feature map $\mathbf{Z}_a$ denoted as value sequence $s = \{s_1, s_2, \ldots, s_L\} \in \mathbb{R}^{1\times L}$.
2: Sort value sequence from large to small and update value sequence $s = \{s_j, s_k, \ldots, s_l\} \in \mathbb{R}^{1\times L}$, where, $s_j > s_k > \ldots > s_l$.
3: Calculate the size of selected band subset according to Eq. (16) and then select the top p values and update value sequence $s = \{s_m, s_n, \ldots, s_q\} \in \mathbb{R}^{1\times p}$ shown in Eq. (14), where, $m < n < \ldots < q$ and then get the optimal bands number sequence $\hat{i} = \{m, n, \ldots, q\} \in \mathbb{R}^{1\times p}$.
4: Achieve the selected band subset $\mathbf{Y} \in \mathbb{R}^{M\times N\times p}$ and corresponding prior-known spectral vector $\hat{\mathbf{d}} \in \mathbb{R}^{1\times p}$ by Eq. (17) and Eq. (18), respectively.

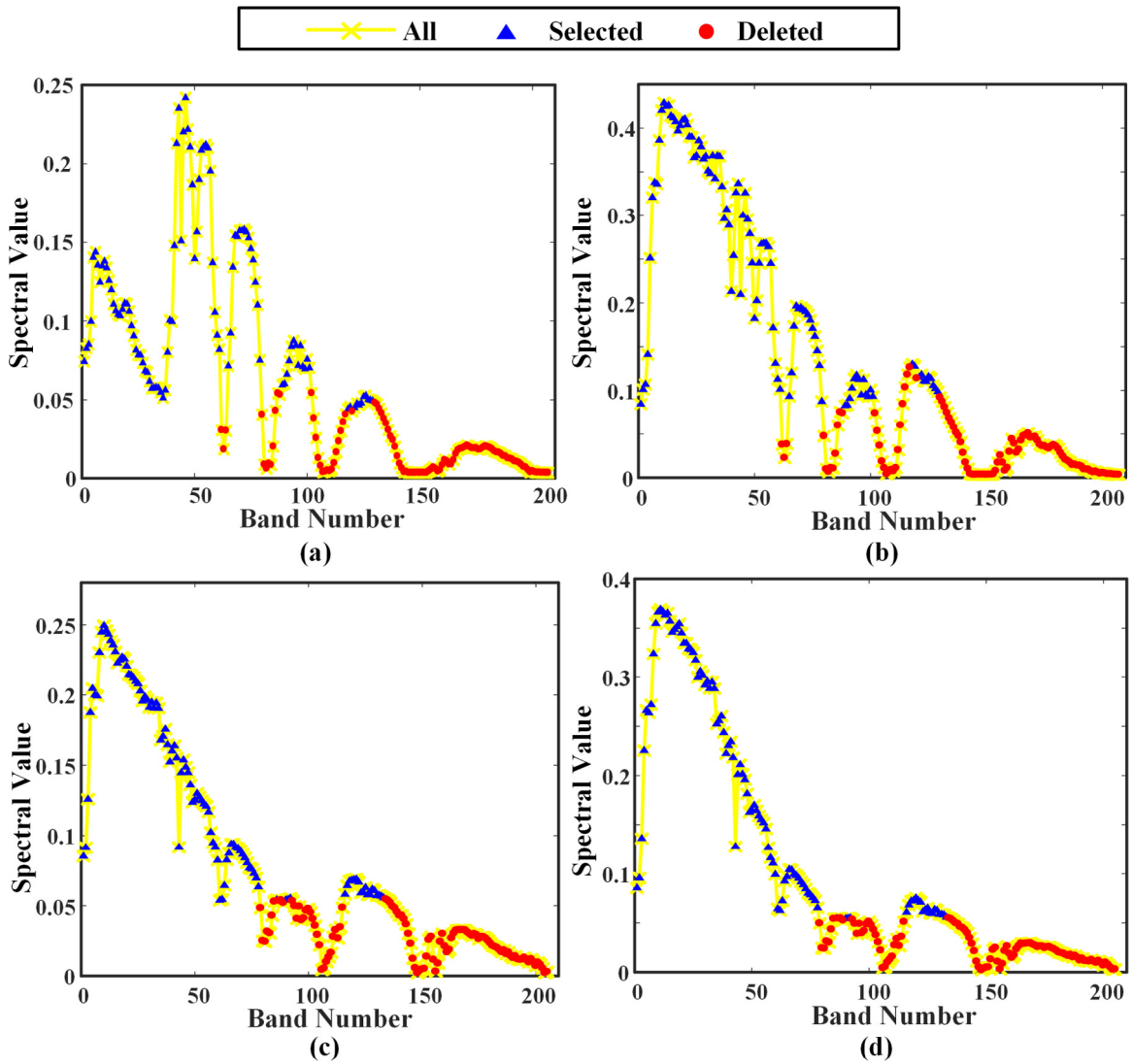

Fig. 4. The results of band selection: spectral curve of one background pixel and one target pixel in (a)(b) for the Texas Coast-2 dataset and (c)(d) for the Los Angels dataset, respectively.

3. Experimental results

3.1. Dataset description

In this section, the proposed method is evaluated on five real hyperspectral datasets (Kang, Zhang, Li, Li, Li, & Benediktsson, 2017) captured at different scenes. Some features of these images are listed in Table 1.

Los Angeles Dataset: It was acquired over the Los Angeles city with a spatial resolution of 7.1 m by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The dataset of Los Angeles contains 100 × 100 pixels and 205 spectral channels ranging from 0.43 to 0.86 um. The houses are regarded as the targets to be detected in this dataset.

Texas Coast Dataset: It is a series of scenes of the Texas coast area captured by the AVIRIS sensor. There are two HSIs of 100 × 100 pixels in this dataset, namely Texas Coast-1 and Texas Coast-2. And the bands of Texas Coast-1 and Texas Coast-2 are 204 and 207, respectively. The spatial resolution of this dataset is 17.2 m, which is corrupted by a series of strip noise and is regarded as a challenging dataset. Buildings with different scales are regarded as targets.

San Diego Dataset: This real HSI is also collected by the AVIRIS sensor over the San Diego airport area, CA, USA, which contains 224 bands spanning from 0.37 to 2.51 um. 189 bands are remained from 224 original bands after eliminating water vapor and atmospheric effect. This data contains 100 × 100 pixels in each band. The targets in this dataset are three airports with 134 pixels.

Pavia Dataset: This real dataset was obtained by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the Pavia city area. It has 100 × 100 pixels and the spatial resolution is 1.3 m. The dataset is comprised of 102 spectral channels from 0.43 to 0.86 um. The kinds of background classes are various, including bridges, rivers, bare soil, and buildings. The vehicles on the bridge are considered targets.

Cuprite Dataset: The dataset was also acquired by the AVIRIS sensor over the Cuprite mining district of Nevada in 1997. Here, a subset of Cuprite data containing 250 × 191 pixels and 188 bands (low-SNR and water absorption bands were removed) is utilized in the experiments. The buddingtonite is considered as the target that should be detected.

Table 1
Some features of the different dataset.

Dataset	Number	Scene	Resolution	Sensor	Fight time	Spatial size	Bands
Los Angeles	1	Airport	7.1 m	AVIRIS	11/9/2011	100 × 100	205
Texas Coast	2	Urban	17.2 m	AVIRIS	8/29/2010	100 × 100	204/207
San Diego	1	Airport	3.5 m	AVIRIS	Unknown	100 × 100	189
Pavia	1	Beach	1.3 m	ROSIS-03	Unknown	100 × 100	102
Cuprite	1	Mining District	Unknown	AVIRIS	Unknown	250 × 191	188

Table 2
The SAM value, AUC and FAR scores of AE, AAE and SSL based detection method.

Date Set	SAM			(TPR,FPR)			(FPR, τ)		
	AE	AAE	SSL	AE	AAE	SSL	AE	AAE	SSL
Los Angeles	0.0517	0.0389	**0.0351**	0.9426	**0.9919**	0.9858	0.0227	**0.0182**	0.0196
Texas Coast-1	**0.0331**	0.0421	0.0549	0.9941	0.9991	**0.9993**	0.0148	0.0117	**0.0083**
Texas Coast-2	0.0781	0.0288	**0.0335**	0.9600	0.9982	**0.9984**	0.1187	0.0176	**0.0051**
San Diego	0.0392	0.0426	**0.0314**	0.9422	0.9880	**0.9933**	**0.0062**	0.0066	0.0066
Pavia	0.2579	0.1645	**0.1540**	0.9895	0.9962	**0.9970**	0.0171	0.0347	**0.0079**
Average	0.0920	0.3395	**0.0618**	0.9650	**0.9947**	**0.9947**	0.0359	0.0180	**0.0095**

3.2. Experimental setup

3.2.1. Comparison methods

To evaluate the effectiveness of the proposed SSTD algorithm, we compare it with five popular hyperspectral target detection algorithms, including CSCR (Li et al., 2015), CEM (Farrand & Harsanyi, 1997), hCEM (Zou & Shi, 2016), ACE (Kraut et al., 2005) and ECEM (Zhao et al., 2019) on real HSIs. Note that these detectors are known as the benchmark and are frequently cited in different kinds of literature. Among these algorithms, CEM and ACE are classical detection algorithms. In the hCEM detector, the traditional CEM detector is used as a basic detector in each layer where the CEM's output of each spectrum is transformed by a nonlinear suppression function and then considered as a coefficient multiplied on this spectrum for the next round of iteration. The deviation δ_k converges to less than 10^{-6} and the parameter λ is restricted to 200 for CEM in our comparative experience. The ECEM is designed based on the classical CEM detection with the help of the ensemble method by integrating several techniques including "cascaded detection", "random averaging" and "multi-scale scanning" to improve both its nonlinear discrimination ability and generalization ability. We set the number of windows and the window size to $n = 4$ and $l_i = \frac{i}{4}L$, $i = 1, \ldots, 4$, where L is the number of bands. The number of detection layers and the number of CEMs per layer is fixed to $k = 10$ and $m = 6$. The CSCR method utilizes combined sparse and collaborative representation. A testing sample is not only sparsely represented by target atoms but also collaboratively represented by background atoms. The decision can be simply achieved by computing the difference between the two representation residuals. But for calculating the false alarm rate, we just adopt the sparse representation for target detection. The parameters λ_1, λ_2 and the window size $(\omega_{in}, \omega_{out})$ are fixed to 10^{-1}, 10^{-2} and $(11, 5)$, respectively.

Each output of these algorithms is scaled to $[0, 1]$ for a fair comparison. In addition, the whole experiments are carried out in Python 3.7.2, TensorFlow-CPU 1.14.0, and MATLAB R2017a environment on a server: Intel(R) Core(TM) i5-7200U CPU at 2.50 GHz.

3.2.2. Evaluation criteria

Evaluation of Reconstruction: Spectral angle mapper (SAM) (Kruse et al., 1993b) is an index to evaluate the spectral similarity between the two input spectral vectors. It can be adopted to measure the evaluation of reconstruction performance. The definition of SAM can be described as:

$$SAM = \frac{1}{m}\sum_{i=1}^{m} \arccos\left(\frac{\mathbf{A}_i\mathbf{B}_i}{\|\mathbf{A}_i\|_2\|\mathbf{B}_i\|_2}\right) \tag{24}$$

where $\mathbf{A}_i$ and $\mathbf{B}_i$ are the two compared spectral vectors and m represents the number of total pixels. In this paper, SAM is restricted at the range of $[0, \pi]$. Moreover, the smaller the value of SAM between the input spectral and the reconstructed spectral, the more similar the two spectral vectors, and the higher the capability of the reconstruction.

Evaluation of Detection: The receiver operating characteristic (ROC) (Zweig & Campbell, 1993) curve and the area under the ROC curve (AUC) (Ferri, Ndezorallo, & Flach, 2011) are adopted to evaluate the ability of the models to recognize targets. The ROC curve describes detection performance with the true positive rate (TPR) and the false positive rate (FPR) at each threshold (τ). The (TPR, FPR) curve describes the tradeoff between TPR and FPR. The (FPR, τ) curve indicates the false alarm rate (FAR) of detection.

3.3. Component and parameter analysis

3.3.1. Component analysis

Effectiveness Evaluation of Reconstruction: The quality of the latent spectral features depends on the reconstruction of the input spectral in the proposed SSL module. The better performance of reconstruction means the better extraction in the deep latent layer. We choose a spectral curve in one pixel of the Pavia dataset whose slight fluctuations in spectral values are more frequent, namely that it is challenging to be trained and learned. As shown in Fig. 5, the SAM between the input spectrum and the reconstructed spectrum is decreased with the increase of iterations. The SSL model enables to fit the curve well gradually after 3000 epochs and reconstruct the original spectral curve adequately in the 6500 epochs. Consequently, the latent units can be considered as a representation of the input spectral vector. Therefore, the feature extraction capacity of our proposed SSL model can be proved.

Effectiveness Evaluation of SSL: To further confirm the necessity of SSL in the proposed method, a component analysis in three circumstances on five datasets is carried out. In the first circumstance, only the AE is used, which is a basic model for joint encoder–decoder training. The second is AAE by adding additional adversarial training D_1 in the hidden layer to AE. For the last one, the data space correction discriminator D_2 is added to the model SSL. We utilize AE and AAE instead of SSL during the comparison experience to detect targets. The SAM values, AUC scores of (TPR, FPR), and FAR scores of (FPR, τ) based on the AE, AAE, and SSL are calculated in Table 2. The AUC scores of the AE-based are low, which means missing some targets, and the detection

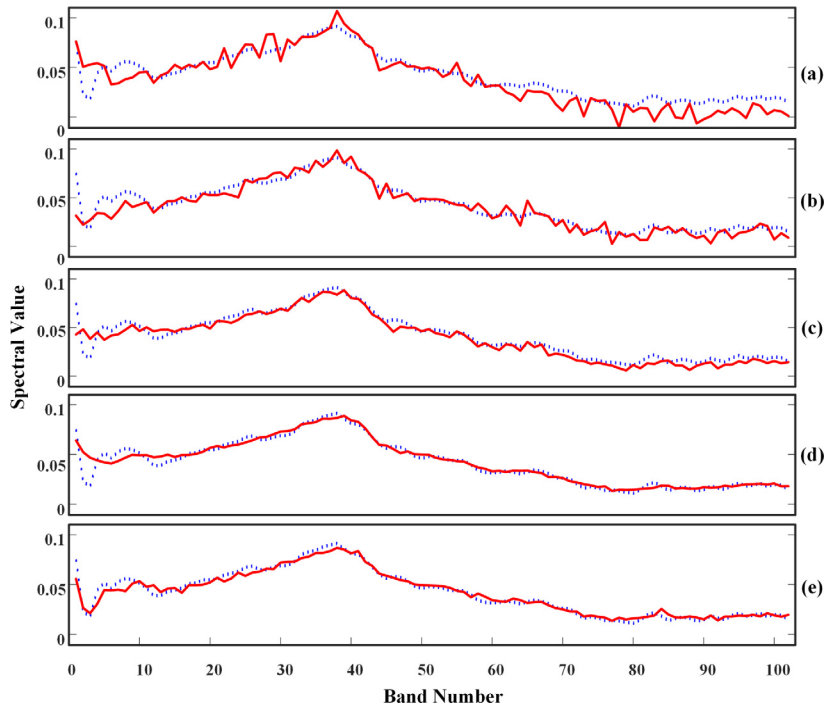

Fig. 5. Reconstructions of the input pixel in different iteration epochs: (a)–(e) reconstructions of input spectrum (blue dotted line) in epoch 100 (SAM=0.22), 1000 (SAM=0.26), 3000 (SAM=0.13), 5000 (SAM=0.11) and 6500 (SAM=0.08), respectively.

performance is worse than SSL in all the datasets. In addition, the detection performance of AAE is better than SSL just on the dataset Los Angeles. The scale of targets in the Los Angeles dataset varies and the number of targets is more than that of the other datasets. As for reconstruction evaluation and detection accuracy, SSL illustrates superior performance on average. In terms of the false alarm rate, the SSL module can obtain lower values on the premise of ensuring detection accuracy. Hence, the SSL module can not only improve the reconstruction effectiveness but also help us procure a better feature matrix in the deep latent space to optimize detection results more than AE and AAE in most cases.

(2) Validation of Band Selection: As shown in Fig. 6, box plots are exploited to validate the necessity and gravity of our band selection algorithm. We arranged an average fusion on the original dataset, selected band subset, and deleted band subset for equal comparison. There are two boxes for each set. The red boxes and green boxes depict the target and the background, respectively. The distance between the two types of boxes reflects the separability between the target and the background. In other words, a large detachable distance means that the set is more suitable and beneficial for detection. It can be recognized that not only the brightness values but also the separation gaps of the selected band subset are larger than the original dataset and much larger than the deleted subset. The former conclusion also can be proved in Fig. 4. These confirm spectral bands obtained by our selection rule become more separable for spectral detection, as informative bands have been emphasized, and less useful ones have been suppressed. This also indirectly supports the powerful discriminability of the hidden layer reference map captured from our SSL module.

(3) Parameter Analysis: This section aims to analyze the effect of different parameters on the performance of the proposed method. The main parameters in this paper are the proportion of selected bands ε and the degree of background suppression δ. ε determines whether the target feature subset is sufficiently selected with redundant information eliminated. As illustrated in Fig. 7(a)(b), when ε equals 0.7, almost all datasets achieve ideal detection performance in terms of detection accuracy and false alarm rate. As discussed in Section 3, the weighted parameter δ balances the influence of the background suppression and detection accuracy enhancement. It is noticed from Fig. 7(c)(d) that when δ equals 5, the detection accuracy and false alarm rate of most datasets are larger than 0.993 and less than 0.01, respectively. Hence, we set the parameter ε and δ as 0.7 and 5, respectively, for all datasets.

3.4. Detection results

The visual detection results of the compared methods on six real hyperspectral datasets are shown in Fig. 8. Fig. 9 shows the ROC curves of (TPR, FPR) on each dataset. It is noticeable that the SSTD is outstanding due to the fact that most of the ROC curves lie nearer the upper left corner.To quantitatively evaluate the performance of the different methods, the AUC, the FAR, and the computing time are illustrated in Table 3, Tables 4 and 5, respectively. It can be observed that the proposed SSTD method offers excellent performance and reaches the highest AUC scores and lower FAR values on average in comparative experience.

For the Los Angeles dataset, as illustrated in Fig. 8(a), the CSCR method fails to detect targets and completely misses the edge shape information of the objections. ACE is capable of detecting a few targets but with a high false alarm rate. CEM, hCEM, and ECEM provide more distinguished detection but lost some targets with background distraction, while the proposed SSTD can effectively preserve the shape of objects with different structures and suppress background with a high detection rate and a low false alarm rate.

For the Texas Coasts dataset, the detection maps are presented in Fig. 8(b)(c). From these maps, it can be observed that CSCR, ACE, CEM, and hCEM can detect the targets with superior accuracy of 0.9929, 0.9593, 0.9409, and 0.9759 in Texas Coast-1, respectively, while these compared algorithms lose some targets in Texas Coast-2. The ECEM produces fewer false alarms; however, it lost a few targets and provided a low accuracy rate of

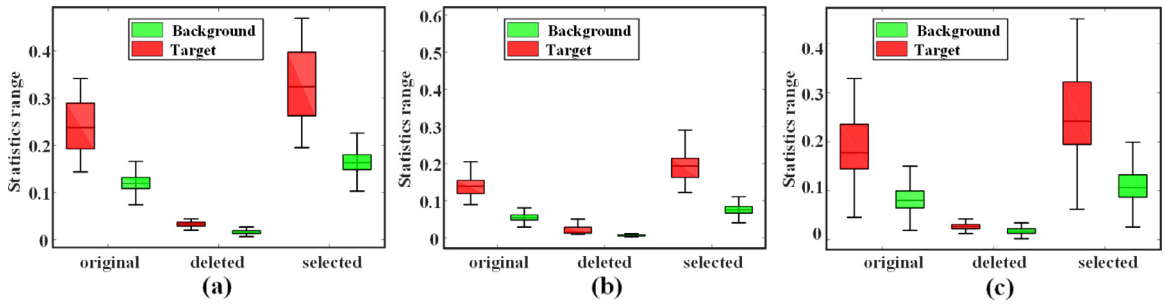

Fig. 6. Background-target separability maps of band selection result for (a) Texas Coast-1, (b) Texas Coast-2, (c) Los Angeles.

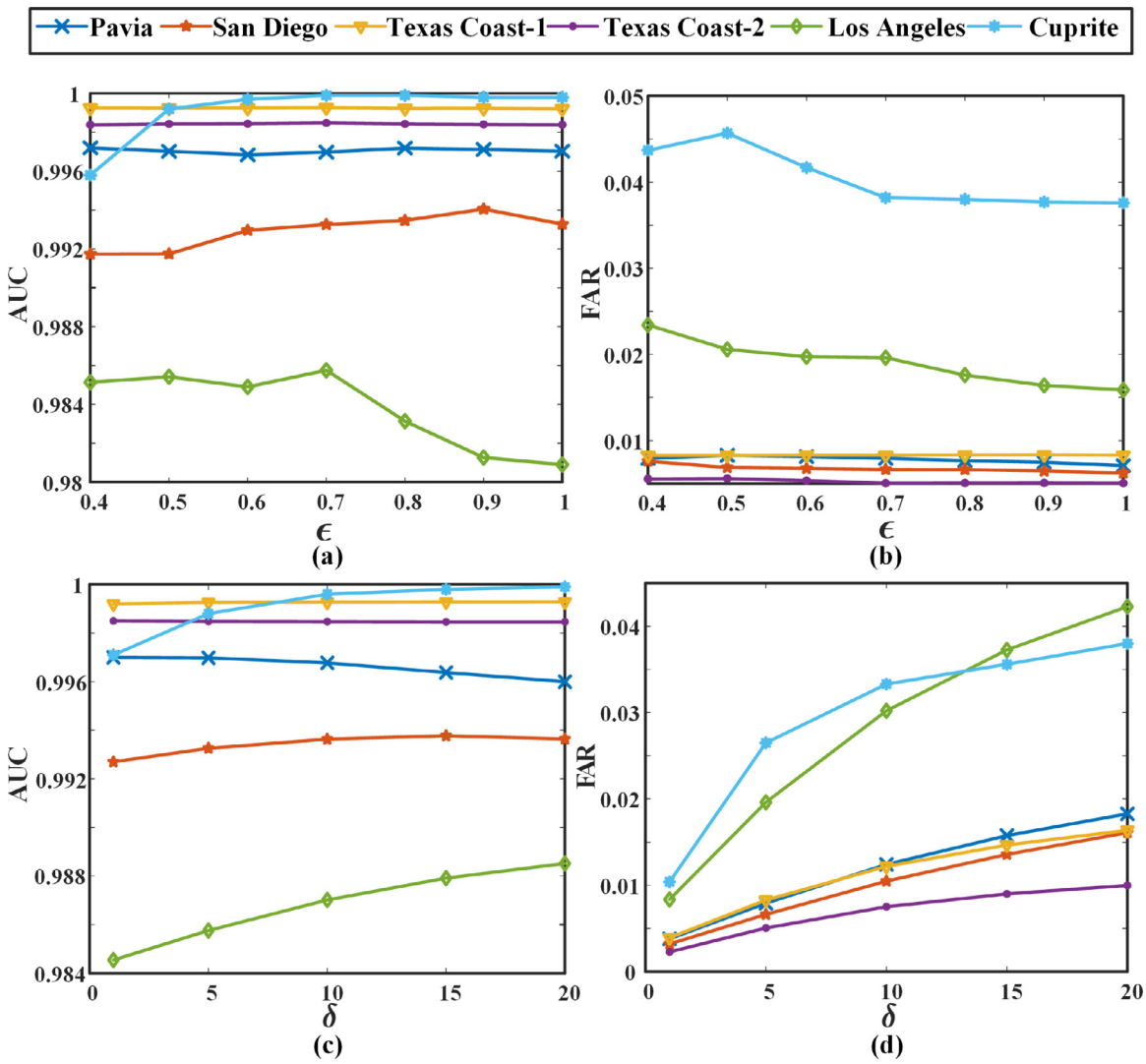

Fig. 7. The sensitivity analysis of the parameters (a)(b) ε and (c)(d) δ on different datasets in AUC and FAR evaluation.

Table 3
The AUC scores obtained by the different methods.

Dataset	SSTD	CSCR	ACE	CEM	hCEM	ECEM
Los Angeles	**0.9858**	0.7003	0.7775	0.7550	0.8488	0.7783
Texas Coast-1	**0.9993**	0.9929	0.9593	0.9409	0.9759	0.8336
Texas Coast-2	**0.9985**	0.7468	0.7361	0.8560	0.8262	0.7565
San Diego	**0.9933**	0.8289	0.9444	0.9560	0.9824	0.9698
Pavia	**0.9970**	0.7420	0.9535	0.9521	0.9750	0.9554
Cuprite	**0.9999**	0.7880	0.9998	**0.9999**	0.9994	0.7073
Average	**0.9956**	0.7496	0.8951	0.9100	0.9087	0.8335

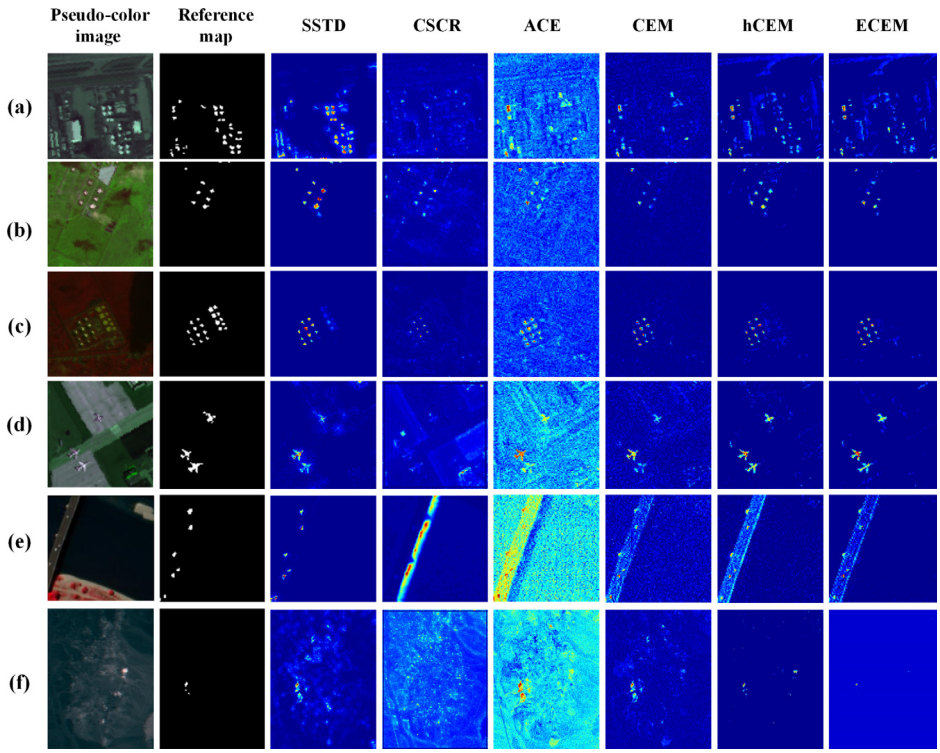

Fig. 8. Pseudo-color image, ground truth, and detection maps of the compared methods for (a) Los Angeles, (b) Texas Coast-1, (c) Texas Coast-2, (d) San Diego, (e) Pavia, (f) Cuprite.

detection. For Texas Coast-1 and Texas Coast-2, the accuracy rate and false alarm rate obtained by SSTD are 0.9993, 0.0083, and 0.9985, 0.0051, respectively. It can be observed that the proposed method, compared with others, provides the optimal detection result, which is more similar to the reference map.

For the San Diego dataset, as shown in Fig. 8(d), ACE and CEM can detect the aircraft, but these two algorithms cannot suppress the background interference. The detection performance of the ECEM algorithm and the hcem algorithm is reliable, avoiding false positives, which are 0.0031 and 0.0027 lower than SSTD, respectively, but the AUC value of SSTD is 0.0160 and 0.0108 higher than SSTD respectively. Besides, the test results of CSCR are not satisfactory.

For the Pavia dataset, Fig. 8(e) shows that the detection result of CSCR misses some targets because of background interference. Ace, CEM, hcem, and ECEM can identify the location of the target, but there are apparent background detection errors. In contrast, SSTD can not only keep the shape of the target but also suppress the background. In particular, the AUC score of the algorithm is 0.9858, which is the highest among all algorithms, and the false alarm rate is 0.0079, which is the lowest among other algorithms, which proves the optimal performance of the algorithm.

For the Cuprite dataset, the detection diagram is shown in the figure. SSTD, CEM, and hcem can detect the target, but it can be seen that the detection result of SSTD is not better than hcem, because its background false detection rate is higher, and the shape of the target is not accurate. Also, the effect of CSCR and EEM algorithm is inferior; both of them cannot locate the high-intensity target.

In addition, the computing time of different methods is calculated to assess computational complexity. Since the introduction of the GAN in the SSL network, the training process of the reconstruction and feature extraction is time-consuming. We only compute the time after the network being learned, and the results of all the detectors are recorded in seconds. As illustrated in Table 5, although the proposed method is not more efficient than the traditional methods, including ACE and CEM, it is much faster in comparison with the CSCR, hCEM, and ECEM. To dig out the main time consumption of the proposed method, we evaluate the computing time of each step. Among them, the adaptive fusion step consumes a relatively large amount of time shown in Table 6. Note that, the average consumption time of step (e) is 262 times shorter than that of the CEM algorithm, which proves that spectral detection can be accelerated after removing redundant information and noise interference through band selection.

4. Conclusion

In this study, a novel SSTD framework with SSL by GAN is proposed for hyperspectral target detection. The main purpose is to learn a relatively discriminative end-to-end feature mapping in deep latent space with background suppression and target saliency. In this connection, three coherent and complementary workflows are presented. Firstly, the SSL module with a spectral GAN is adopted to offer the discriminative feature matrix between target and background in the deep latent space. Subsequently, a spatial detector is employed on the adaptive weighted feature map. As for spectral consideration, an improved CEM is conducted on the selected band subset along with the corresponding selected target signature. Especially, a novel band selection rule in an unsupervised way is constructed by considering the structure, contrast, and luminance similarity between the discriminative feature matrix in latent space and the original full band in data space. Finally, an adaptive fusion combines the complementary information in different aspects, followed by an exponential constrained nonlinear transform function to suppress the background. Experimental results on the various real HSIs suggest that our SSTD algorithm surpasses some typical or creative methods such as CEM, hCEM, ACE, ECEM, and CSCR in terms of quantitative evaluation, visual comparison, and computational efficiency. In the future, we will pay more attention to the multi-scale method and try to reduce the computation time by designing a more efficient adaptive fusion step.

Table 4
The false alarm rate obtained by the different methods.

Dataset	SSTD	CSCR	ACE	CEM	hCEM	ECEM
Los Angeles	0.0079	0.0406	0.2124	0.0316	0.0161	**0.0155**
Texas Coast-1	0.0083	0.0382	0.1832	0.0150	0.0017	**0.0013**
Texas Coast-2	0.0051	0.0183	0.1698	0.0119	**0.0014**	0.0015
San Diego	0.0066	0.0615	0.2723	0.0420	0.0039	**0.0035**
Pavia	**0.0196**	0.0511	0.3726	0.0638	0.0246	0.0164
Cuprite	0.0382	0.1893	0.3381	0.0411	**0.0001**	0.0805
Average	0.0143	0.0686	0.2581	0.0342	**0.0125**	0.0198

Table 5
The computing time (in seconds) of the different methods.

Dataset	SSTD	CSCR	ACE	CEM	hCEM	ECEM
Los Angeles	0.4047	245.8804	0.0966	**0.0435**	1.3833	6.9494
Texas Coast-1	0.4067	242.6438	0.1009	**0.0426**	1.0824	6.7450
Texas Coast-2	0.4383	249.3246	0.1042	**0.0627**	1.0773	7.2137
San Diego	0.4107	190.9867	0.1263	**0.0504**	1.1519	6.1216
Pavia	0.2715	57.1730	0.0848	**0.0323**	1.1098	4.6935
Cuprite	1.6105	950.7661	0.4926	**0.2412**	6.2923	4.1335
Average	0.5904	322.7958	0.1676	**0.0788**	2.0162	5.9761

Table 6
Time consumption (in seconds) of each step: Step(a): Adaptive fusion, Step(b): Attribute filter, Step(c): Guide filter, Step(d): Band selection, Step(e): Spectral detector, Step(f): Spatial–spectral fusion and Step(g): Background suppression.

DataSet	SSTD							CEM
	Step(a)	Step(b)	Step(c)	Step(d)	Step(e)	Step(f)	Step(g)	
Los Angeles	0.2211	0.0861	0.3065	0.0636	**0.0002**	0.0283	0.0006	0.0435
Texas Coast-1	0.2295	0.0823	0.3097	0.0538	**0.0002**	0.02744	0.0006	0.0426
Texas Coast-2	0.2105	0.080557	0.3084	0.0546	**0.0002**	0.0263	0.0007	0.0627
San Diego	0.2151	0.1042	0.2934	0.0437	**0.0003**	0.0217	0.0008	0.0504
Pavia	0.247	0.112	0.1914	0.0349	**0.0002**	0.0215	0.0006	0.0323
Cuprite	0.4612	0.2409	0.722	0.1503	**0.0006**	0.0616	0.0013	0.2412
Average	0.2641	0.1177	0.3552	0.0668	**0.0003**	0.0311	0.0008	0.0788

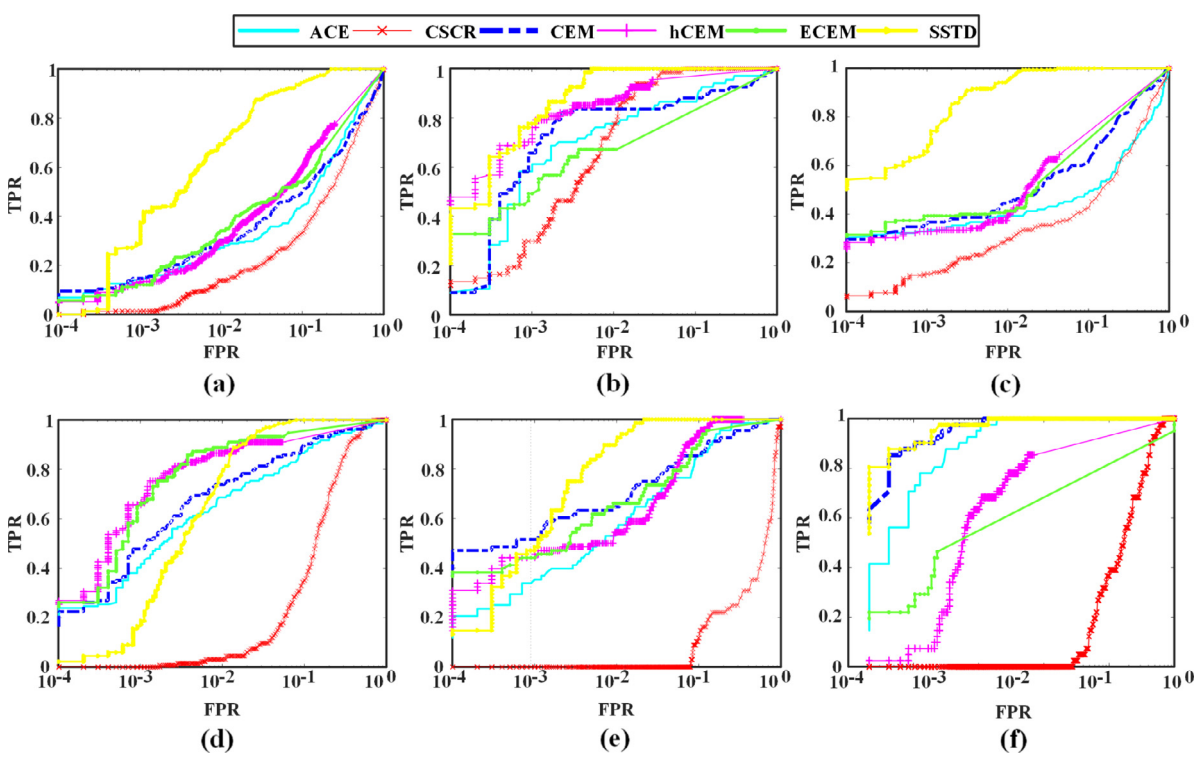

Fig. 9. ROC curves of the SSTD method for (a) Los Angeles, (b) Texas Coast-1, (c) Texas Coast-2, (d) Sandiego, (e) Pavia, (f) Cuprite.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grants 62071360, 61801359, 61571345, 91538101, 61501346, 61502367, and 61701360, in part by the Young Talent fund of the University Association for Science and Technology in Shaanxi of China under Grant 20190103, in part by the Special Financial Grant from the China Postdoctoral Science Foundation under Grant 2019T120878, in part by the 111 projects under Grant B08038, in part by the Fundamental Research Funds for the Central Universities, China under Grant JB180104, in part by the Natural Science Basic Research Plan in Shaanxi Province of China under 2019JQ153, Grant 2016JQ6023, and Grant 2016JQ6018, in part by the General Financial Grant from the China Postdoctoral Science Foundation under Grant 2017M620440, in part by the Yangtse Rive Scholar Bonus Schemes, China under Grant CJT160102, and in part by the Ten Thousand Talent Program, China.

References

Badrinarayanan, V., Kendall, A., & Cipolla, R. (2015). SegNet: A deep convolutional encoder-decoder architecture for image segmentation. ArXiv E-Prints ArXiv:1511.00561.

Bitar, A. W., Cheong, L., & Ovarlez, J. (2019). Sparse and low-rank matrix decomposition for automatic target detection in hyperspectral imagery. *IEEE Transactions on Geoscience and Remote Sensing*, *57*(8), 5239–5251.

Breen, E. J., & Jones, R. (1996). Attribute openings, thinnings, and granulometries. *Computer Vision and Image Understanding*, *64*(3), 377–389.

Cai, Y., Liu, X., & Cai, Z. (2020). BS-Nets: An end-to-end framework for band selection of hyperspectral image. *IEEE Transactions on Geoscience and Remote Sensing*, *58*(3), 1969–1984.

Cao, V. L., Nicolau, M., & Mcdermott, J. (2016). A hybrid autoencoder and density estimation model for anomaly detection. In *Proceedings of 14th international conference on parallel problem solving from nature* (pp. 717–726).

Charles, A. S., Olshausen, B. A., & Rozell, C. J. (2011). Learning sparse codes for hyperspectral imagery. *IEEE Journal of Selected Topics in Signal Processing*, *5*(5), 963–978.

Chein-I Chang (2000). An information-theoretic approach to spectral variability, similarity, and discrimination for hyperspectral image analysis. *IEEE Transaction on Information Theory*, *46*(5), 1927–1932.

Chen, Y., Jiang, H., Li, C., Jia, X., & Ghamisi, P. (2016). Deep feature extraction and classification of hyperspectral images based on convolutional neural networks. *IEEE Transactions on Geoscience and Remote Sensing*, *54*(10), 6232–6251.

Chen, Y., Nasrabadi, N. M., & Tran, T. D. (2011). Sparse representation for target detection in hyperspectral imagery. *IEEE Journal of Selected Topics in Signal Processing*, *5*(3), 629–640.

Dalla Mura, M., Atli Benediktsson, J., Waske, B., & Bruzzone, L. (2010). Morphological attribute profiles for the analysis of very high resolution images. *IEEE Transactions on Geoscience and Remote Sensing*, *48*(10), 3747–3762.

Farrand, W. H., & Harsanyi, J. C. (1997). Mapping the distribution of mine tailings in the Coeur d'Alene River Valley, Idaho, through the use of a constrained energy minimization technique. *Remote Sensing of Environment*, *59*(1), 64–76.

Feng, J., Li, D., Chen, J., Zhang, X., Tang, X., & Wu, X. (2019). Hyperspectral band selection based on ternary weight convolutional neural network. In *2019 IEEE international geoscience and remote sensing symposium* (pp. 3804–3807).

Ferri, C. S., Ndezorallo, J. H., & Flach, P. A. (2011). A coherent interpretation of AUC as a measure of aggregated classification performance. In *Proceedings of the 28th international conference on machine learning* (pp. 657–664).

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., et al. (2014). Generative adversarial networks. ArXiv Preprint ArXiv:1406.2661.

Gu, J., Wang, Z., Kuen, J., Ma, L., Shahroudy, A., Shuai, B., et al. (2015). Recent advances in convolutional neural networks. ArXiv E-Prints ArXiv:1512.07108.

Harsanyi, J. C., & Chang, C. (1994). Hyperspectral image classification and dimensionality reduction: an orthogonal subspace projection approach. *IEEE Transactions on Geoscience and Remote Sensing*, *32*(4), 779–785.

Hinton, G. E., & Zemel, R. S. (1994). Autoencoders, minimum description length and Helmholtz free energy. *Advances in Neural Information Processing Systems*, *6*.

Ioffe, S., & Szegedy, C. (2015). Batch normalization: Accelerating deep network training by reducing internal covariate shift. ArXiv Preprint ArXiv:1502.03167.

Jiang, T., Li, Y., Xie, W., & Du, Q. (2020). Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection. *IEEE Transactions on Geoscience and Remote Sensing*, *58*(7), 4666–4679.

Kang, X., Zhang, X., Li, S., Li, K., Li, J., & Benediktsson, J. A. (2017). Hyperspectral anomaly detection with attribute and edge-preserving filters. *IEEE Transactions on Geoscience and Remote Sensing*, *55*(10), 5600–5611.

Kelly, E. J. (1986). An adaptive detection algorithm. *IEEE Transactions on Aerospace and Electronic Systems*, *AES-22*(2), 115–127.

Kraut, S., Scharf, L. L., & Butler, R. W. (2005). The adaptive coherence estimator: a uniformly most-powerful-invariant adaptive detection statistic. *IEEE Transactions on Signal Processing*, *53*(2), 427–438.

Kraut, S., Scharf, L. L., & McWhorter, L. T. (2001). Adaptive subspace detectors. *IEEE Transactions on Signal Processing*, *49*(1), 1–16.

Kruse, F. A., Lefkoff, A. B., Boardman, J. W., Heidebrecht, K. B., Shapiro, A. T., Barloon, P. J., et al. (1993a). The spectral image processing system (SIPS) interactive visualization and analysis of imaging spectrometer data. *Remote Sensing of Environment*, *44*(2), 145–163.

Kruse, F. A., Lefkoff, A. B., Boardman, J. W., Heidebrecht, K. B., Shapiro, A. T., Barloon, J. P., et al. (1993b). The spectral image processing system (sips): Interactive visualization and analysis of imaging spectrometer data. *Remote Sensing of Environment*, *44*(2–3), 145–163.

Li, W., Du, Q., & Zhang, B. (2015). Combined sparse and collaborative representation for hyperspectral target detection. *Pattern Recognition*, *48*(12), 3904–3916.

Li, W., Wu, G., & Du, Q. (2017). Transferred deep learning for anomaly detection in hyperspectral imagery. *IEEE Geoscience and Remote Sensing Letters*, *14*(5), 597–601.

Liu, X., Wang, C., Wang, H., Fu, M., Feng, Y., Bourennane, S., et al. (2019). Target detection of hyperspectral image based on faster R-CNN with data set adjustment and parameter turning. In *OCEANS 2019 - Marseille* (pp. 1–8).

Lockwood, R. B., Cooley, T., Jacobson, J., & Manolakis, D. G. (2009). Is there a best hyperspectral detection algorithm?. *Proceedings of SPIE*, *7334*, 733402–733402–16.

Ma, X., Wang, H., & Geng, J. (2016). Spectral spatial classification of hyperspectral image based on deep auto-encoder. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *9*(9), 4073–4085.

Makhzani, A., Shlens, J., Jaitly, N., Goodfellow, I., & Frey, B. (2015). Adversarial autoencoders. ArXiv Preprint ArXiv:1511.05644.

Manolakis, D., & Shaw, G. (2002). Detection algorithms for hyperspectral imaging applications. *IEEE Signal Processing Magazine*, *19*(1), 29–43.

Nasrabadi, N. M. (2008). Regularized spectral matched filter for target recognition in hyperspectral imagery. *IEEE Signal Processing Letters*, *15*, 317–320.

Redmon, J., & Farhadi, A. (2018). YOLOv3: An incremental improvement. ArXiv E-Prints ArXiv:1804.02767.

Robey, F. C., Fuhrmann, D. R., Kelly, E. J., & Nitzberg, R. (1992). A CFAR adaptive matched filter detector. *IEEE Transactions on Aerospace and Electronic Systems*, *28*(1), 208–216.

Scharf, L. L., & Friedlander, B. (1993). Matched subspace detectors. *1*, In *Proceedings of 27th asilomar conference on signals, systems and computers* (pp. 701–706).

Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, *13*(4), 600–612.

Xie, W., Jiang, T., Li, Y., Jia, X., & Lei, J. (2019). Structure tensor and guided filtering-based algorithm for hyperspectral anomaly detection. *IEEE Transactions on Geoscience and Remote Sensing*, *57*(7), 4218–4230.

Xie, W., Lei, J., Yang, J., Yang, Y., Du, Q., & Li, Z. (2020). Deep latent spectral representation learning-based hyperspectral band selection for target detection. *IEEE Transactions on Geoscience and Remote Sensing*, *58*(3), 2015–2026.

Xie, W., Li, Y., Lei, J., Yang, J., Chang, C., & Li, Z. (2020). Hyperspectral band selection for spectral spatial anomaly detection. *IEEE Transactions on Geoscience and Remote Sensing*, *58*(5), 3426–3436.

Xie, W., Yang, J., Lei, J., Li, Y., Du, Q., & He, G. (2020). SRUN: Spectral regularized unsupervised networks for hyperspectral target detection. *IEEE Transactions on Geoscience and Remote Sensing*, *58*(2), 1463–1474.

Yang, S., & Shi, Z. (2016). Hyperspectral image target detection improvement based on total variation. *IEEE Transactions on Image Processing*, *25*(5), 2249–2258.

Zare, A., Jiao, C., & Glenn, T. (2018). Discriminative multiple instance hyperspectral target characterization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *40*(10), 2342–2354.

Zhai, M., Liu, H., & Sun, F. (2019). Lifelong learning for scene recognition in remote sensing images. *IEEE Geoscience and Remote Sensing Letters*, *16*(9), 1472–1476.

Zhan, Y., Tian, H., Liu, W., Yang, Z., Wu, K., Wang, G., et al. (2017). A new hyperspectral band selection approach based on convolutional neural network. In *2017 IEEE international geoscience and remote sensing symposium* (pp. 3660–3663).

Zhang, M., Gong, M., Mao, Y., Li, J., & Wu, Y. (2019). Unsupervised feature extraction in hyperspectral images based on wasserstein generative adversarial network. *IEEE Transactions on Geoscience and Remote Sensing*, *57*(5), 2669–2688.

Zhang, L., Zhang, L., Tao, D., Huang, X., & Du, B. (2014). Hyperspectral remote sensing image subpixel target detection based on supervised metric learning. *IEEE Transactions on Geoscience and Remote Sensing*, *52*(8), 4955–4965.

Zhao, R., Shi, Z., Zou, Z., & Zhang, Z. (2019). Ensemble-based cascaded constrained energy minimization for hyperspectral target detection. *Remote Sensing*, *11*(11), 1310.

Zou, Z., & Shi, Z. (2016). Hierarchical suppression method for hyperspectral target detection. *IEEE Transactions on Geoscience and Remote Sensing*, *54*(1), 330–342.

Zou, Z., & Shi, Z. (2016). Hierarchical suppression method for hyperspectral target detection. *IEEE Transactions on Geoscience and Remote Sensing*, *54*(1), 330–342.

Zweig, M. H., & Campbell, G. (1993). Receiver-operating characteristic (ROC) plots: a fundamental evaluation tool in clinical medicine. *Clinical Chemistry*, *39*(4), 561–577.